

Giulia Merlini

Postdoctoral researcher
CERMICS - ENPC

Born in 1996
Italian nationality
Paris, France
(+33) 06 11 29 70 22
giulia.merlini@enpc.fr



Education and Career

Research positions

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| 2025 - | Postdoctoral researcher at CERMICS - MATHÉRIALS team, at ENPC (France).
Supervisors: <i>C. Le Bris</i> and <i>F. Legoll</i> .
Subject: Development of efficient computational methods for multiscale hyperbolic conservation laws. |
| 2021 - 2025 | Ph.D. at LMS - MÈDISIM team, Ecole Polytechnique (France).
Supervisors: <i>J.-M. Allain</i> and <i>S. Imperiale</i> .
Subject: Mechanical and numerical modelling of transient shear wave elastography for the cornea. |
| 2020 - 2021 | Master Internship at LMS - MÈDISIM team, Ecole Polytechnique, France.
Supervisors: <i>C. Giraudet</i> and <i>P. Le Tallec</i> .
Subject: Patient specific models with incomplete and mixed data: the example of the human cornea. |

Education

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| 2021 - 2025 | Ph.D. in Fluid and Solid Mechanics, Acoustics , Ecole Polytechnique (France).
Members of jury: <i>O. Doaré</i> (president), <i>B. Lombard</i> (rapporteur and examiner), <i>J. Diaz</i> (rapporteur and examiner), <i>A. Pandolfi</i> (examiner), <i>A. Niclas</i> (examiner), <i>G. Rosi</i> (examiner). |
| 2018 - 2021 | M.Sc. degree in Biomechanics and Biomaterials , Biomedical engineering, at Politecnico di Milano (Italy). |
| 2019 | Erasmus program at Universitat Politècnica de València, (Spain). |
| 2015 - 2018 | B.Sc. degree in Automation engineering , at Politecnico di Milano (Italy). |

Scientific Dissemination

Articles

1. **Merlini, G.**, Allain, J.-M., Imperiale, S. Fully explicit numerical scheme for linearized wave propagation in nearly-incompressible soft hyperelastic solids. *Wave Motion*, 2025, Volume 139, 27 pages. Q2 (0.570). DOI : <https://doi.org/10.1016/j.wavemoti.2025.103594>.
2. **Merlini, G.**, Imperiale, S., Allain, J.-M. Finite element modeling of transient shear wave elastography for the cornea. *Journal of the Mechanics and Physics of Solids*, 2025, Volume 206, Part A, 22 pages. Q1 (1.757). DOI : <https://doi.org/10.1016/j.jmps.2025.106363>.

Oral presentations in international conferences

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| 2026 | | HYP , 20th International Conference on Hyperbolic Problems: Theory, Numerics and Applications, Stuttgart (Germany). |
| 2025 | | SIMAI , Biennial congress of the Italian Society of Applied and Industrial Mathematics, SISSA, Trieste (Italy). |
| 2025 | | COLIBRI , Focus Workshop 2025 on Computational Medicine, Graz (Austria). |
| 2024 | | WAVES , 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation, Berlin, (Germany). |
| 2023 | | CMBBE , 18th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering, Paris (France). |

Oral and poster presentations in national conferences and seminars

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| 2026 | | Invited talk , XVIIeme Colloque Franco-Roumain de Mathématiques Appliquées, Nancy. |
| 2026 | | Invited seminar , MAP5, Université Paris Cité, Paris. |
| 2024 | | JCJC Ondes 2024 , MAP5, Université Paris Cité, Paris. |
| 2024 | | Guided waves: from theory to experiment , Poster presentation at Laboratoire MAP5, Université Paris Cité, Paris. |
| 2023 | | SB Congress , 48eme Congrès de la Société Biomécanique, Grenoble. |
| 2023 | | F2M-msp , Journée Thématique Biomécanique et Biomatériaux, Ecole Polytechnique, Palaiseau. |
| 2022 | | Journée IDEFIX/MEDISIM/POEMS , Poster presentation at INRIA Saclay, Palaiseau. |

Social initiatives

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| 2025 - ongoing | | Programme Étudiants Réfugiés Volunteering in math teaching for refugee students |
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Teaching

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| 2025 | | Tutorials (TD), ENPC, Undergrad level, 8h/year
Modelling with partial differential equations |
| 2024 | | Tutorials , Ecole Polytechnique, M2 level, 8h/year
Mechanics of deformable continuum |
| 2024 | | Tutorials , Ecole Polytechnique, M2 level, 28h/year
Biomechanics in Health and Disease |
| 2023 | | Tutorials (TD), Bachelor of Ecole Polytechnique, L1 level, 28h/year
Physics 1 |
| 2022 - 2023 | | Tutorials (PC), Ecole Polytechnique, Undergrad level, 64h/year
Modal in mechanics |

Skills

Languages

- Italian (mother tongue)
- English (C1)
- French (C1)
- Spanish (B2)

Code

- C++: Contributed to the development of a library for 3D elastic wave simulations
- Matlab: use for scientific production
- Python: use for scientific and teaching production
- FreeFem++: use for scientific production